

Patterns of cancer incidence among US Hispanics/Latinos, 1995–2000

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Abstract

Objective Current and comprehensive data on cancer incidence in US Latinos has been limited.

Methods Using a standardized approach to uniformly assign Hispanic/Latino race/ethnicity to cancer records, data from 15 central cancer registries, representing more than 85% of the US Latino population, were included in the analysis. Average annual age-adjusted incidence rates and standard errors were calculated for Hispanic, non-Hispanic white and non-Hispanic black males and females. To compare cancer incidence among Hispanic and non-Hispanic populations, standardized incidence ratios (SIRs) also were calculated.

Results Latino populations had overall lower incidence for all cancers combined and the four leading cancers (breast, prostate, lung and colorectal) than non-Hispanic populations, however, cancers of lesser impact in non-Hispanic populations (liver, gallbladder, stomach, penis and cervix) were more commonly diagnosed among Latinos.

Conclusions Understanding the patterns of cancer incidence in this diverse racial/ethnic minority group can serve to both stimulate research into the unique

behaviors, exposures and genetics that drive cancer risk among Latinos and to direct prevention and control efforts tailored to this population.

Keywords Cancer incidence · Hispanic/Latino populations

Introduction

The 2000 US census clearly confirmed the Hispanic/Latino population as the nation's fastest growing minority group. While the US population has grown by 13% since the 1990 census, the Latino population has grown by 58% and now comprises approximately 13% of the total US population, a proportion similar to that of African-Americans [1]. One of the challenges to providing cancer incidence data for this population has been the lack or inconsistency of methods used by population-based cancer registries to identify Latinos among reported cancer cases. Methods used to assign Hispanic/Latino ethnicity have included one or a combination of the following approaches: medical record notation, birthplace information, death certificates, race, visual cues or identification, self-identification in research studies used to supplement registry information, and use of surnames and maiden names to identify Latino heritage. Many registries did not use any protocol and did not produce information about Latino populations in their registry [2].

Therefore, in order to address the need for reliable and comprehensive national cancer incidence data for Latinos, it is necessary to first develop a standardized approach to Latino identification in cancer registry

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records. To that end, the North American Association of Central Cancer Registries (NAACCR), a professional organization that, among other activities, develops uniform data standards for cancer registration and certifies population-based registries for data quality for incidence statistics, convened an expert panel to develop a standard Hispanic/Latino identification protocol for use by its member registries. This protocol was applied by all registries responding to a special NAACCR call for data to produce Latino cancer incidence rates [3]. This report is based on these NAACCR data, which represent the first comprehensive national cancer incidence rates for the US Hispanic/Latino population.

Materials and methods

Participating registries

In April 2003, NAACCR issued a call for data from its US member registries, requesting they submit cancer data for diagnosis years 1995 through 2000 for the purpose of generating national cancer incidence statistics for Hispanic/Latino, non-Hispanic white and non-Hispanic black populations. Fifteen population-based central cancer registries, including those with the largest Latino populations, met the NAACCR criteria for data completeness and quality (i.e. silver or gold certification levels) for inclusion in the combined cancer statistics for Latino populations presented in this report. Fourteen state central cancer registries (Arizona, California, Colorado, Florida, Idaho, Illinois, Iowa, New Jersey, New Mexico, New York, North Carolina, Texas, Washington, and Wyoming) participated, as did one metropolitan area (Detroit, Michigan). The data from these registries were used for the comparison white and black non-Hispanic populations as well.

NAACCR best practice guideline for identification of Hispanic/Latino persons

Hispanic ethnicity was assigned to the cancer cases through the standardized use of the NAACCR Hispanic Identification Algorithm (NHIA), the details of which are available in the final report of the NAACCR Expert Panel on Hispanic Identification [4]. Briefly, NHIA uses a combination of NAACCR variables, including Spanish/Hispanic origin, birthplace, race, surname, and maiden name, to directly or indirectly classify cases as Hispanic/Latino for analytic purposes. Cases reported as having Spanish/Hispanic

origin (as indicated by a standard NAACCR data element reported by all member registries) are directly identified as Hispanic in the dataset. Cases reported as non-Spanish/non-Hispanic, Spanish by surname only, or unknown whether Spanish are evaluated for possible Hispanic ethnicity by the algorithm through indirect identification. The ultimate goal of the algorithm is to classify these cases as Hispanic or non-Hispanic based on an evaluation of the strength of the birthplace, race, and/or surname associations with Hispanic ethnicity status. After the individual registries have applied NHIA to their database, cases not ultimately classified as Hispanic are classified as non-Hispanic, leaving no cases with “unknown” Hispanic status, and these data are reported to NAACCR. Hispanic ethnicity is assigned regardless of race of the individual, e.g., a case having a race code of black and a positive indicator for Hispanic status according to the algorithm would be assigned Hispanic ethnicity.

Statistical analysis

Rates for Hispanic, non-Hispanic white and non-Hispanic black males and females were calculated per 100,000 population and age-adjusted by the direct method to the 2000 US population standard. Hispanic rates presented include all races, however, only a small number of Hispanic cases had a race code other than white. The incidence rates are annual averages for the period 1995 through 2000. Estimates of the population used for the calculation of rates for the US for 1995 through 2000 were obtained from the National Cancer Institute (NCI) Surveillance, Epidemiology and End Results (SEER) program, based on US Bureau of Census population estimates for these years. These population estimates represent a modification of the annual time series of population estimates produced by the Population Estimates Program of the Bureau of the Census with support from the NCI [5]. The standard error of the difference between adjusted cancer incidence rates was used to calculate a Z (standard normal) statistic in order to evaluate the statistical significance of rate differences among the populations [6]. To further quantify the difference in incidence rates between Hispanic/Latino and non-Hispanic males and females, standardized incidence ratios (SIRs) were calculated by dividing the age-adjusted incidence rate for Hispanics by the age-adjusted incidence rate for non-Hispanic white and non-Hispanic black populations. Corresponding 95% confidence intervals for the SIRs also were calculated.

Results

A total of 3,645,670 cancers (all invasive and bladder in situ) were diagnosed in the 6 year study period (1995–2000) among the central cancer registries comprising the NAACCR US combined areas. Of these, 306,918 (8.4%) were diagnosed among Hispanic/Latino populations, 2,998,532 (82.2%) among non-Hispanic white populations, and 340,220 (9.3%) among non-Hispanic black populations. The areas covered in the NAACCR combined cancer incidence rates used in this study represent more than 85% of the total US Hispanic/Latino population, however, only 45.2% of the US non-Hispanic white population and 47.2% of the US non-Hispanic black population is covered by the registries providing these data.

Common cancers in Hispanic/Latino populations

The leading cancers among Latino males were prostate (27.2% of all cancers among Latino males), lung and bronchus (11.4%), colorectal (11.1%), non-Hodgkin lymphoma (5.6%), and bladder (4.1%). These top five sites accounted for 59.4% of the total cancers diagnosed among Latino males. The leading cancers among Latina females were breast (29.2% of all cancers among Latina females), colorectal (9.9%), lung and bronchus (6.8%), cervix uteri (6.2%), and corpus uteri (5.2%). Together, these sites accounted for 57.3% of total cancers diagnosed among Latina females.

Comparison of common cancer sites with non-Hispanic populations

Total cancers

For the majority of cancer sites, average annual age-adjusted cancer incidence rates were statistically significantly lower among Hispanic males and females than their non-Hispanic counterparts (Tables 1–3). The overall rate for cancer among Latino males was 443.0 per 100,000, compared to 562.4 per 100,000 among non-Hispanic white males and 661.0 per 100,000 among non-Hispanic black males. Among Latina females, the overall cancer rate was 313.0 per 100,000, compared to 434.9 among non-Hispanic white females and 391.7 per 100,000 among non-Hispanic black females.

Prostate cancer

Although this was the leading type of cancer diagnosed in Latino males, the incidence rate was significantly lower than the corresponding rate in non-Hispanic

white (SIR = 0.88) and non-Hispanic black males (SIR = 0.55).

Lung and bronchus cancer

This cancer was among the top five cancers diagnosed in both Hispanic males and females; however, the magnitude of the incidence rates was approximately half that of the rates in their non-Hispanic counterparts, a difference that was statistically significant.

Breast cancer (in females)

Breast cancer was the most commonly diagnosed cancer in Latina females, with an incidence rate of 89.2 per 100,000. Although a leading cancer, Latina breast cancer rates were statistically significantly less than those of both non-Hispanic white females (SIR = 0.64) and non-Hispanic black females (SIR = 0.79).

Colon cancer

Colon cancer was the second most common cancer diagnosed among Latina females (26.3 per 100,000) and the third most common cancer diagnosed among Latino males (35.7 per 100,000). Incidence rates for both Hispanic males and females were substantially lower, however, than those of non-Hispanic males and females, a difference that was statistically significant.

Cervical cancer

The incidence rate for cervical cancer was 1.8 times higher among Latina females than non-Hispanic white females, a statistically significant difference, but was equivalent to the incidence rate among non-Hispanic black females.

Bladder cancer

With the exception of non-Hispanic black males (SIR = 1.17, 95% CI = 1.12, 1.22), Hispanic incidence rates for bladder cancer were statistically significantly lower than those in non-Hispanic populations.

Non-Hodgkin lymphoma (NHL)

A common cancer in Hispanics, NHL incidence rates for Latino males (20.1 per 100,000) and Latina females (13.6 per 100,000) were statistically significantly lower than their non-Hispanic white counterparts, but slightly higher than those for non-Hispanic black males

Table 1 Incidence rates^a for Hispanic, non-Hispanic white and non-Hispanic black males and females. Selected areas in the US, 1995–2000

Cancer site	Males						Females					
	Hispanic		Non-Hispanic white		Non-Hispanic black		Hispanic		Non-Hispanic white		Non-Hispanic black	
	Rate	SE	Rate	SE	Rate	SE	Rate	SE	Rate	SE	Rate	SE
All sites	443.0	1.25	562.4	0.46	661.0	1.64	313.0	0.84	434.9	0.36	391.7	0.99
Oral cavity and pharynx	12.4	0.20	16.8	0.08	20.7	0.27	4.0	0.10	6.8	0.05	6.4	0.13
Lip	0.9	0.06	2.0	0.03	0.1	0.02	0.2	0.02	0.4	0.01	0.1	0.01
Tongue	2.7	0.09	4.0	0.04	4.5	0.12	1.0	0.05	1.7	0.02	1.3	0.06
Major salivary gland	1.1	0.06	1.7	0.03	1.1	0.07	0.7	0.04	0.9	0.02	0.9	0.05
Floor of mouth	1.0	0.05	1.3	0.02	2.0	0.08	0.2	0.02	0.6	0.01	0.5	0.04
Gum and other mouth	1.8	0.08	2.2	0.03	3.1	0.10	0.9	0.05	1.4	0.02	1.4	0.06
Nasopharynx	0.7	0.04	0.7	0.02	1.2	0.06	0.2	0.02	0.3	0.01	0.4	0.03
Oropharynx	0.6	0.05	0.6	0.01	1.3	0.07	0.1	0.02	0.2	0.01	0.3	0.03
Hypopharynx	1.5	0.07	1.5	0.02	3.0	0.10	0.2	0.02	0.4	0.01	0.5	0.04
Digestive system	108.4	0.63	109.9	0.20	139.0	0.76	69.2	0.42	73.2	0.14	92.9	0.49
Esophagus	6.0	0.15	7.8	0.05	13.5	0.23	1.5	0.06	2.1	0.02	4.0	0.10
Stomach	17.3	0.26	10.2	0.06	19.9	0.30	9.0	0.15	4.4	0.04	10.2	0.16
Small intestine	1.4	0.07	1.8	0.03	2.9	0.11	1.1	0.05	1.3	0.02	2.0	0.07
Colon excluding rectum	35.7	0.37	47.1	0.13	53.8	0.48	26.3	0.26	35.9	0.10	43.0	0.34
Rectum and rectosigmoid junction	17.1	0.24	19.6	0.09	17.3	0.26	9.0	0.15	11.9	0.06	11.4	0.17
Anus	1.0	0.05	1.2	0.02	1.4	0.07	1.3	0.06	1.6	0.02	1.3	0.06
Liver	12.4	0.20	5.2	0.04	9.5	0.19	4.6	0.11	1.9	0.02	3.2	0.09
Gallbladder	1.6	0.08	0.8	0.02	1.0	0.07	3.3	0.09	1.3	0.02	1.8	0.07
Pancreas	11.4	0.21	12.7	0.07	16.3	0.26	9.2	0.15	9.7	0.05	13.5	0.19
Respiratory system	66.7	0.50	104.8	0.20	135.5	0.74	26.3	0.25	61.1	0.13	52.7	0.37
Larynx	7.5	0.16	8.1	0.05	13.1	0.22	1.0	0.05	1.8	0.02	2.5	0.08
Lung and bronchus	56.7	0.46	93.5	0.19	120.1	0.70	24.5	0.25	58.2	0.13	49.4	0.36
Bones and joints	1.0	0.05	1.2	0.02	0.9	0.05	0.8	0.04	0.9	0.02	0.7	0.04
Soft tissue including heart	3.1	0.09	3.6	0.04	3.4	0.11	2.4	0.07	2.6	0.03	2.7	0.08
Skin excluding basal and squamous	7.7	0.15	25.8	0.10	5.3	0.12	4.4	0.10	15.4	0.07	2.2	0.07
Melanoma of the skin	3.9	0.11	22.3	0.09	1.1	0.07	3.4	0.08	14.2	0.07	0.9	0.05
Breast (excl. in situ)	1.0	0.06	1.5	0.02	2.0	0.09	89.2	0.44	138.7	0.21	113.2	0.53
Female genital system	–	–	–	–	–	–	49.3	0.32	55.3	0.13	51.0	0.35
Cervix uteri	–	–	–	–	–	–	16.0	0.17	8.9	0.06	15.6	0.19
Corpus uteri	–	–	–	–	–	–	16.1	0.19	24.0	0.09	18.1	0.21
Uterus, NOS	–	–	–	–	–	–	0.9	0.05	0.8	0.02	1.6	0.06
Ovary	–	–	–	–	–	–	13.0	0.17	17.9	0.08	12.0	0.17
Male genital system	140.2	0.71	160.3	0.24	247.5	1.01	–	–	–	–	–	–
Prostate	135.2	0.70	152.8	0.24	245.0	1.01	–	–	–	–	–	–
Testis	3.4	0.06	6.4	0.05	1.2	0.05	–	–	–	–	–	–
Penis	1.4	0.07	0.8	0.02	1.0	0.07	–	–	–	–	–	–
Urinary system	37.6	0.37	59.6	0.15	35.5	0.39	14.3	0.19	19.4	0.08	16.0	0.20
Bladder (incl. in situ)	21.2	0.29	41.8	0.13	18.1	0.29	5.4	0.12	10.6	0.05	7.1	0.14
Kidney, renal pelvis	15.5	0.22	16.4	0.08	16.7	0.25	8.6	0.14	8.2	0.05	8.4	0.15
Ureter	0.5	0.05	1.0	0.02	0.3	0.04	0.2	0.03	0.4	0.01	0.2	0.02
Eye	0.6	0.04	1.1	0.02	0.3	0.03	0.4	0.03	0.8	0.02	0.2	0.02
Brain and other nervous system	6.1	0.13	8.9	0.06	4.8	0.12	4.7	0.10	6.4	0.05	3.7	0.09
Brain	5.7	0.12	8.4	0.06	4.3	0.12	4.3	0.09	5.9	0.04	3.2	0.09
Endocrine system	3.3	0.09	4.5	0.04	2.7	0.09	9.5	0.13	10.2	0.06	6.1	0.12
Thyroid	2.7	0.08	3.8	0.04	1.9	0.08	9.0	0.13	9.6	0.06	5.4	0.11
Lymphomas	23.0	0.26	26.8	0.10	20.6	0.26	15.6	0.19	19.0	0.08	13.0	0.18
Hodgkin's disease	2.9	0.08	3.4	0.04	2.8	0.08	2.0	0.06	2.8	0.03	1.9	0.06
Non-Hodgkin's lymphoma	20.1	0.25	23.4	0.09	17.8	0.25	13.6	0.18	16.3	0.07	11.1	0.16
Multiple myeloma	6.4	0.15	6.2	0.05	12.3	0.23	4.5	0.10	4.0	0.03	9.7	0.16
Leukemias	11.9	0.19	16.5	0.08	12.1	0.22	7.8	0.13	9.7	0.05	7.8	0.14
Acute lymphocytic	2.2	0.05	1.7	0.03	1.0	0.05	1.7	0.05	1.3	0.02	0.7	0.04
Chronic lymphocytic	2.4	0.10	5.2	0.04	3.8	0.13	1.3	0.06	2.7	0.03	1.9	0.07
Acute myeloid	3.5	0.11	4.8	0.04	3.5	0.12	2.5	0.07	3.1	0.03	2.6	0.08
Chronic myeloid	2.0	0.08	2.1	0.03	1.9	0.08	1.2	0.05	1.2	0.02	1.2	0.05

Table 1 continued

Cancer site	Males						Females					
	Hispanic		Non-Hispanic white		Non-Hispanic black		Hispanic		Non-Hispanic white		Non-Hispanic black	
	Rate	SE	Rate	SE	Rate	SE	Rate	SE	Rate	SE	Rate	SE
Other	0.9	0.06	1.4	0.02	1.2	0.07	0.7	0.04	0.8	0.02	0.9	0.05
Ill-defined & unknown	13.6	0.23	14.9	0.08	18.3	0.28	10.6	0.16	11.4	0.06	13.5	0.19

^aIncidence rates are per 100,000 population and are age-adjusted using the 2000 US population standard

Data source: North American Association of Central Cancer Registries file submissions as of July 2003 from Arizona, California, Colorado, Florida, Idaho, Illinois, Iowa, New Jersey, New Mexico, New York, North Carolina, Texas, Washington, Wyoming and metropolitan Detroit, Michigan

Table 2 Standardized incidence ratios comparing Hispanic males to non-Hispanic white and non-Hispanic black males. Selected areas of the US, 1995–2000

Rank for Hispanics	Cancer site	Standardized incidence ratio (95% CI) ^a			
		Hispanic:non-Hispanic white		Hispanic:non-Hispanic black	
	All sites	0.79	(0.78, 0.79)	0.67	(0.67, 0.68)
1	Prostate	0.88	(0.88, 0.89)	0.55	(0.54, 0.56)
2	Lung and bronchus	0.61	(0.60, 0.62)	0.47	(0.46, 0.48)
3	Colon excl. rectum	0.76	(0.74, 0.77)	0.66	(0.65, 0.68)
4	Bladder (incl. in situ)	0.51	(0.49, 0.52)	1.17	(1.12, 1.22)
5	Non-Hodgkin's lymphoma	0.86	(0.84, 0.88)	1.13	(1.09, 1.17)
6	Stomach	1.70	(1.64, 1.75)	0.87	(0.83, 0.91)
7	Rectum and rectosigmoid	0.87	(0.85, 0.90)	0.99	(0.95, 1.03)
8	Kidney and renal pelvis	0.95	(0.92, 0.97)	0.93	(0.89, 0.97)
9	Oral cavity and pharynx	0.74	(0.71, 0.76)	0.60	(0.58, 0.62)
10	Liver	2.38	(2.30, 2.47)	1.31	(1.24, 1.37)
11	Leukemia	0.72	(0.70, 0.75)	0.98	(0.94, 1.03)
12	Pancreas	0.90	(0.86, 0.93)	0.70	(0.67, 0.73)
13	Multiple myeloma	1.03	(0.98, 1.08)	0.52	(0.49, 0.55)
14	Brain and other nervous system	0.69	(0.66, 0.72)	1.27	(1.19, 1.36)
15	Esophagus	0.77	(0.73, 0.81)	0.44	(0.42, 0.47)
16	Melanoma of the skin	0.17	(0.17, 0.18)	3.55	(3.09, 4.06)
17	Testis	0.53	(0.51, 0.55)	2.83	(2.59, 3.10)
18	Hodgkin's disease	0.85	(0.80, 0.90)	1.04	(0.96, 1.12)
19	Thyroid	0.71	(0.67, 0.76)	1.42	(1.28, 1.57)
20	Gallbladder	2.00	(1.79, 2.23)	1.60	(1.35, 1.89)
21	Penis	1.75	(1.57, 1.95)	1.40	(1.18, 1.66)

^a95% CI: lower bound, upper bound of 95% confidence interval ; Standard incidence ratio (SIR) confidence interval calculation: Exponentiate: $\ln R_H - \ln R_{NH} \pm z1.96\sqrt{(SE(R_H)/R_H)^2 + (SE(R_{NH})/R_{NH})^2}$ where R_H is the rate for Hispanic populations and R_{NH} is the rate for either non-Hispanic white or non-Hispanic black populations

Data source: North American Association of Central Cancer Registries file submissions as of July 2003 from Arizona, California, Colorado, Florida, Idaho, Illinois, Iowa, New Jersey, New Mexico, New York, North Carolina, Texas, Washington, Wyoming and metropolitan Detroit, Michigan

(SIR = 1.13, 95% CI = 1.09, 1.17) and non-Hispanic black females (SIR = 1.23, 95% CI = 1.18, 1.27).

Stomach cancer

The incidence rate of stomach cancer among Latino males was 1.7 times that of non-Hispanic white males, a

statistically significant difference, but was slightly lower than the rate for non-Hispanic black males (SIR = 0.87, 95% CI = 0.83, 0.91). As with males, the stomach cancer incidence rate for Latina females was twice as high as the non-Hispanic white female rate, again a statistically significant difference, and slightly lower than the non-Hispanic black female rate (SIR = 0.88, 95% CI = 0.84, 0.92).

Table 3 Standardized incidence ratios comparing Hispanic females to non-Hispanic white and non-Hispanic black females. Selected areas of the US, 1995–2000

Rank for Hispanics	Cancer site	Standardized incidence ratio (95% CI)			
		Hispanic:non-Hispanic white		Hispanic:non-Hispanic black	
	All sites	0.72	(0.72, 0.72)	0.80	(0.79, 0.80)
1	Breast (excl. in situ)	0.64	(0.64, 0.65)	0.79	(0.78, 0.80)
2	Colon excluding rectum	0.73	(0.72, 0.75)	0.61	(0.60, 0.63)
3	Lung and bronchus	0.42	(0.41, 0.43)	0.50	(0.48, 0.51)
4	Corpus uteri	0.67	(0.65, 0.69)	0.89	(0.86, 0.92)
5	Cervix uteri	1.80	(1.75, 1.84)	1.03	(0.99, 1.06)
6	Non-Hodgkin's lymphoma	0.83	(0.81, 0.86)	1.23	(1.18, 1.27)
7	Ovary	0.73	(0.71, 0.75)	1.08	(1.04, 1.13)
8	Pancreas	0.95	(0.92, 0.98)	0.68	(0.65, 0.71)
9	Stomach	2.05	(1.97, 2.12)	0.88	(0.84, 0.92)
10	Rectum and rectosigmoid	0.76	(0.73, 0.78)	0.79	(0.76, 0.82)
11	Thyroid	0.94	(0.91, 0.97)	1.67	(1.59, 1.75)
12	Kidney and renal pelvis	1.05	(1.01, 1.09)	1.02	(0.98, 1.07)
13	Leukemias	0.80	(0.78, 0.83)	1.00	(0.95, 1.05)
14	Bladder (incl. in situ)	0.51	(0.49, 0.53)	0.76	(0.72, 0.81)
15	Brain and other nervous system	0.73	(0.70, 0.77)	1.27	(1.19, 1.35)
16	Liver	2.42	(2.30, 2.55)	1.44	(1.34, 1.55)
17	Multiple myeloma	1.13	(1.07, 1.18)	0.46	(0.44, 0.49)
18	Oral cavity and pharynx	0.59	(0.56, 0.62)	0.63	(0.59, 0.67)
19	Melanoma of the skin	0.24	(0.23, 0.25)	3.78	(3.36, 4.25)
20	Gallbladder	2.54	(2.39, 2.70)	1.83	(1.67, 2.01)
21	Hodgkin's disease	0.71	(0.67, 0.76)	1.05	(0.97, 1.15)
22	Esophagus	0.71	(0.66, 0.77)	0.38	(0.34, 0.41)

^a95% CI: lower bound, upper bound of 95% confidence interval ; Standard incidence ratio (SIR) confidence interval calculation:

Exponentiate: $\ln R_H - \ln R_{NH} \pm 1.96 \sqrt{(\text{SE}(R_H)/R_H)^2 + (\text{SE}(R_{NH})/R_{NH})^2}$ where R_H is the rate for Hispanic populations and R_{NH} is the rate for either non-Hispanic white or non-Hispanic black populations

Data source: North American Association of Central Cancer Registries file submissions as of July 2003 from Arizona, California, Colorado, Florida, Idaho, Illinois, Iowa, New Jersey, New Mexico, New York, North Carolina, Texas, Washington, Wyoming and metropolitan Detroit, Michigan

Liver cancer

The liver cancer incidence rate among Latino males was 2.4 times higher than among non-Hispanic white males, and 1.3 times higher than among non-Hispanic black males, both statistically significant differences. Similarly, the liver cancer rate for Latina females was statistically significantly higher than the rate for non-Hispanic white and non-Hispanic black females (SIRs = 2.42 and 1.44, respectively).

Gallbladder cancer

The incidence rate for gallbladder cancer among Latino males was 2.0 times higher than the non-Hispanic white male rate, and 1.6 times higher than the non-Hispanic black male rate, both statistically significant differences. Incidence of gallbladder cancer also was statistically significantly greater in Latina females, with their incidence being 2.5 times higher than the non-Hispanic white female rate, and 1.8 times higher than the non-Hispanic black female rate.

Testicular cancer

The incidence rate for cancer of the testis among Latino males (3.4 per 100,000) was approximately half that of non-Hispanic white males (6.4 per 100,000), however, it was 2.8 times that of non-Hispanic black males (1.2 per 100,000), with both differences being statistically significant.

Penile cancer

The incidence rate for cancer of the penis among Latino males was 1.8 times higher than non-Hispanic white males and 1.4 times higher than non-Hispanic black males, with both differences being statistically significant.

Ovarian cancer

The rate of ovarian cancer was statistically significantly lower among Latina women (13.0 per 100,000)

compared to non-Hispanic white women (17.9 per 100,000), but only slightly higher than the rate for non-Hispanic black women (12.0 per 100,000).

Uterine cancer

Latina women had a statistically significantly lower rate of uterine cancer (16.1 per 100,000) than both non-Hispanic white (24.0 per 100,000) and non-Hispanic black women (18.1 per 100,000).

Cancers of the buccal cavity and pharynx

The rate of cancers of the buccal cavity and pharynx was statistically significantly lower among the Hispanic population than among the non-Hispanic population, regardless of gender.

Discussion

These data indicate that cancer incidence for all cancer sites combined is substantially lower among Latino males and Latina females than among non-Hispanic white and non-Hispanic black males and females, due largely to lower incidence of prostate, lung and bronchus, breast, and colon cancers. However, for specific cancers including gallbladder, liver, stomach, cervical, penile, testicular, thyroid, and brain cancers and NHL, there are patterns of increased incidence among Hispanic populations as compared to non-Hispanic white and/or non-Hispanic black populations. These differing patterns of incidence for total and specific cancers have been noted previously in studies of the epidemiology of cancer in Hispanic populations; however, the comparisons have been mainly to non-Hispanic white populations [7–10]. Seeing several distinct patterns when comparing the cancer experience of Hispanic populations to non-Hispanic black as well as non-Hispanic white populations (i.e., Hispanic rates higher than non-Hispanic rates; Hispanic rates higher than non-Hispanic white but equivalent to non-Hispanic black; Hispanic rates intermediate between non-Hispanic white and non-Hispanic black) gives added dimension to considering the etiology of and prevention strategies for these cancers in these populations.

One pattern seen in these data is the increased incidence of liver and gallbladder cancers in Latino males and Latina females and of penile cancer in Latino males. Although the greatest excess for these cancers is seen in comparison to non-Hispanic white males and females, Hispanic rates are also higher than those of non-Hispanic black males and females. A

variety of tumors occur in the liver, but the most commonly diagnosed type is hepatocellular carcinoma (HCC). Unlike many cancers, there are well-established risk factors for HCC, including chronic infection with hepatitis B and C viruses, alcohol consumption and cigarette smoking, dietary exposure to aflatoxins, low intake of antioxidant vitamins and selenium, high intake of iron, long-term ingestion of inorganic arsenic and occupational exposure to thorotrast and vinyl chloride [11, 12]. For gallbladder cancer among North American populations in general and Mexican Americans in particular, the single strongest risk factor identified to date is a history of cholesterol gallstones, a condition more prevalent in women than men. However, the risk factors for these stones (obesity, multiple pregnancies, low plasma high density lipoprotein (HDL), female hormones and insulin resistance) do not account for all observed incidence, leaving a possible role for genetic susceptibility. In addition, there is accumulating evidence for a role for chronic infection in the development of gallbladder cancer [13, 14]. Studies of the etiology of penile cancer generally implicate sexual transmission of an infectious agent, with human papillomavirus (HPV) a likely candidate [15]. The elevated rates of both penile and cervical cancers are consistent with that hypothesis. Higher rates of liver, gallbladder and penile cancers in Hispanics are likely due to a higher prevalence in this population of one or more of the identified risk factors.

A second interesting pattern seen in these data was that for stomach and cervical cancers. For these cancers, incidence rates are higher in Latinos than in non-Hispanic white populations but lower or equivalent to those in non-Hispanic black populations. In recent years, biological as well as epidemiological evidence has clearly established the role of HPV as the sexually transmitted agent in the etiology of cervical cancer, to the point of meriting public health action, possibly through HPV testing and vaccination in high risk populations [16, 17]. Similarly, an infectious agent, *Helicobacter pylori*, has been implicated in the etiology of stomach cancers within the last decade [18, 19]. Higher infection rates for both HPV and *H. pylori* have been noted in black and Hispanic populations when compared to white populations [20, 21].

A final pattern noted in these data was for those cancer sites where the highest rate was seen among non-Hispanic white males and females, the lowest rate was among non-Hispanic black males and females, and an intermediate rate was seen for Hispanic males and females. The cancers exhibiting this incidence pattern included bladder, brain, melanoma, testicular, thyroid and NHL. Risk factors for these cancers, where known,

are quite diverse and include a mix of environmental and occupational factors [22–26]. Interestingly, however, there is a consistent socio-economic gradient seen, with higher income associated with higher risk, for brain, NHL and testicular cancers; this gradient may also exist for thyroid cancers, but is less well established.

While NHIA is the best standardized method available to date for identifying Hispanic ethnicity indirectly in population-based cancer surveillance programs, some limitations should be noted. The indirect method of the algorithm relies heavily on the Spanish surname list from the 1990 census as well as birthplace and other information (such as maiden name) recorded in the medical record. In general, use of surname without additional information such as self-reported ethnicity, maiden name, or birthplace may result in misclassification. Use of the surname list may not be appropriate for populations in areas of the US that were once occupied by Spain and have adopted Hispanic surnames but are not actually Hispanic by inheritance. Similarly, someone with a reported birthplace of a country of largely Hispanic population may not personally have a Hispanic heritage. Additionally, the surname portion of the algorithm has been found to be overly sensitive in some counties having sparse Hispanic populations. NHIA has since been modified such that registries can tailor the surname portion to address this problem for counties that meet a set threshold for percent Hispanic population. The magnitude of the misclassification resulting from these scenarios, however, is likely to be small and should not overly impact the interpretation of these data. One purpose of studying cancer incidence by race/ethnicity is to reveal differences in disease risk among distinct populations and, ultimately, to identify risk factors associated with different cultures. NHIA aids in this goal by capturing individuals living in a heavily Hispanic environment as the ones of interest when estimating the cancer burden of Hispanics.

Despite some potential limitations in the algorithm, the use of a standardized approach to the identification of Hispanic/Latino individuals in population-based cancer registry data for a large proportion of the US Hispanic/Latino population is a strength of this study. The size of the Hispanic/Latino population included in the study area should result in more reliable estimates of cancer incidence than has been available previously. As these data suggest, the pattern of cancer incidence among the Hispanic population differs sharply from that of other racial groups in the United States. Cancer prevention among the Hispanic population will require culturally-relevant strategies that can address a

complex mix of environmental, cultural and socio-economic risk factors. Of particular interest is the higher incidence among Hispanic men and women of cancers having an infectious etiology. This could be a reflection of either the impact of recent immigrants on this population or the overall lower socio-economic conditions of this segment of the US population—or both. The picture will be clearer, and more complex, when the uniform NAACCR algorithm for assigning Hispanic ethnicity to population-based cancer incidence data begins to provide national cancer statistics for Hispanic subgroups with potentially unique etiologies (e.g. Mexican, Cuban, South and Central American, Puerto Rican) and for Hispanic populations in differing regions of the United States.

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